

Singapore Airlines Review Analysis

**Specialist Diploma in Applied
Artificial Intelligence
Capstone Project 2021**



Problem Statement

- Singapore airlines receives a large volume of customer reviews containing valuable feedback about passengers' experiences.
- However, manually analysing these reviews can be time-consuming and may make it difficult to identify common concerns, positive experiences, and patterns in customer feedback.
- Therefore, text analytics techniques are needed to systematically analyse the reviews and uncover meaningful insights into customer sentiment, key topics, and different groups of customer experiences.



Project Objectives

1. **Analyze customer sentiment** in Singapore Airlines reviews to identify positive or negative feedback and determine the main words associated with each sentiment.
2. **Identify key topics and patterns** in reviews using topic modelling and clustering to understand the main areas of customer experience.
3. **Develop a text classification model** using sentiment, dominant topic, and cluster as features to classify customer satisfaction into low, neutral, and high rating categories.

What is Sentiment Analysis (VADER)?



VADER (Valence Aware Dictionary and Sentiment Reasoner) is a lexicon- and rule- based sentiment analysis method, particularly effective for social media and informal text.



Produces **four** sources: **Positive, Negative, Neutral, and Compound.**



Compound score interpretation commonly used:

$\geq 0.05 \rightarrow$ Positive
 $\leq -0.05 \rightarrow$ Negative
 -0.05 to $0.05 \rightarrow$ Neutral

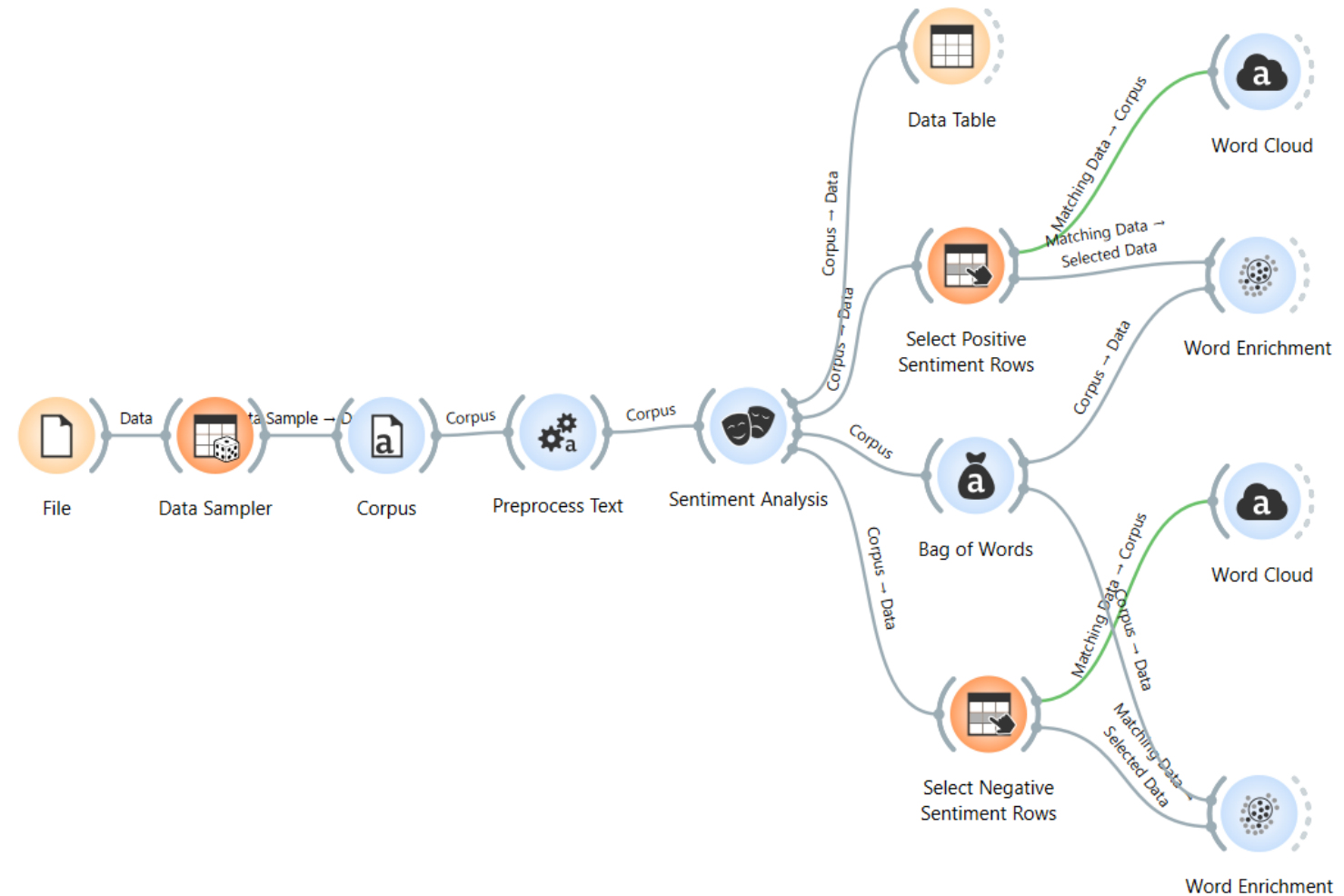


Advantages: Fast, simple, no training data required, good for short/ informal text.



Limitations: May struggle with sarcasm, domain-specific terminology, and complex content.

Sentiment Analysis Pipelines



- The pipeline imports and samples the review data, preprocesses the text, performs sentiment analysis, and generates separate word clouds and word enrichment results for positive and negative reviews.

Sentiment Analysis Word Cloud & Enrichment

Positive Sentiment



Word Enrichment - Orange

Info
Words displayed: 37

Filter
☒ p-value 0.0500
☒ FDR 0.2000

Word	p-value	FDR
great	1.1e-15	2.6e-13
friendly	9.8e-13	1.2e-10
comfortable	9.2e-12	7.2e-10
excellent	1.5e-10	8.6e-09
entertainment	8.1e-09	3.8e-07
good	6.6e-08	2.6e-06
best	1.6e-07	5.2e-06
always	6.6e-06	2.0e-04
highly	9.0e-06	2.3e-04

Send Automatically

740 | 1000 37

Negative Sentiment



Word Enrichment - Orange

Info
Words displayed: 89

Filter
☒ p-value 0.0500
☒ FDR 0.2000

Word	p-value	FDR
ticket	2.2e-10	5.2e-08
customer	1.8e-09	2.1e-07
never	9.1e-09	7.2e-07
told	5.9e-08	3.3e-06
poor	7.0e-08	3.3e-06
disappointed	1.4e-07	5.4e-06
asked	2.0e-07	6.8e-06
call	8.1e-07	2.4e-05
day	1.7e-06	4.5e-05
later	1.9e-06	4.5e-05

Send Automatically

118 | 1000 89

Insights from Sentiment Analysis



Positive Sentiment: Reviews frequently mention friendly service, comfortable experiences, excellent service, good food, and entertainment. Words such as great, friendly, comfortable, and excellent are significantly enriched.



Negative Sentiment: Negative reviews focus more on tickets, customer service, poor experiences, and complaints, with enriched words such as ticket, customer, poor, disappointed, and asked. Issues also relate to food, seats, flights, and service.



Overall, positive reviews emphasise Singapore Airlines service quality and comfortable travel experience, while negative reviews mainly highlight customer service issues and problems with tickets and the overall flight experience.

What is Hierarchical Clustering?

Hierarchical clustering groups similar data into a hierarchy of clusters.

Euclidean Distance: Measures the straight line distance between two data points.

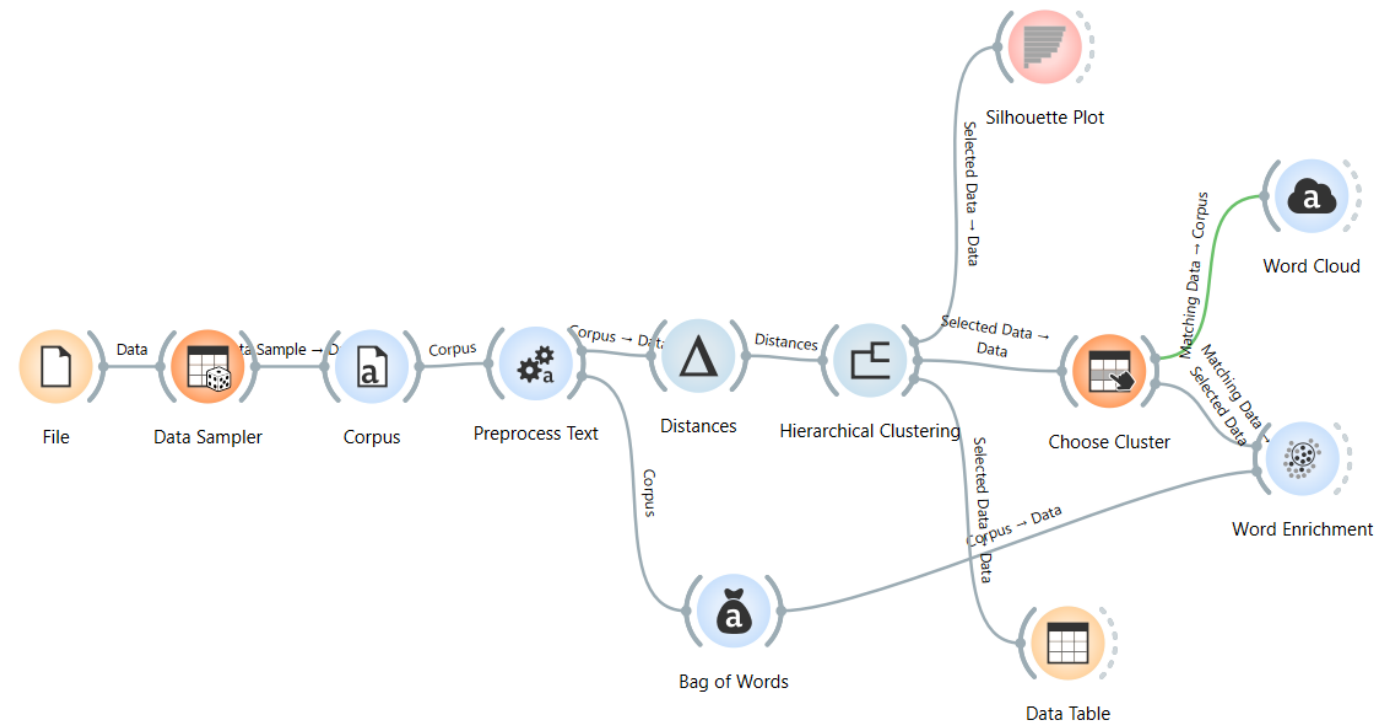
- Smaller distances – data points are more similar
- Larger distances – data points are less similar

Agglomerative Process:

- Start with each data point as an individual cluster.
- Calculate the Euclidean distance between clusters.
- Merge the closest clusters.
- Repeat until all clusters form a hierarchy.
- Use a dendrogram to determine the final clusters.

Clustering Analysis Pipelines

- The pipeline preprocesses the review text, converts it into word representations, performs hierarchical clustering, evaluates the clusters using a silhouette plot, and analyses the selected clusters through word clouds and word enrichment.



Clustering Analysis Word Cloud

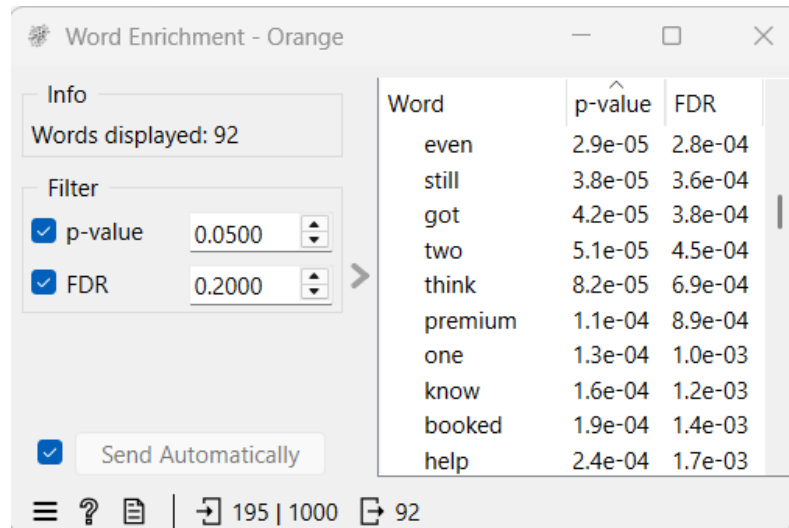
C1

C2

C3

Clustering Analysis Word Enrichment

C1



Word Enrichment - Orange

Info
Words displayed: 92

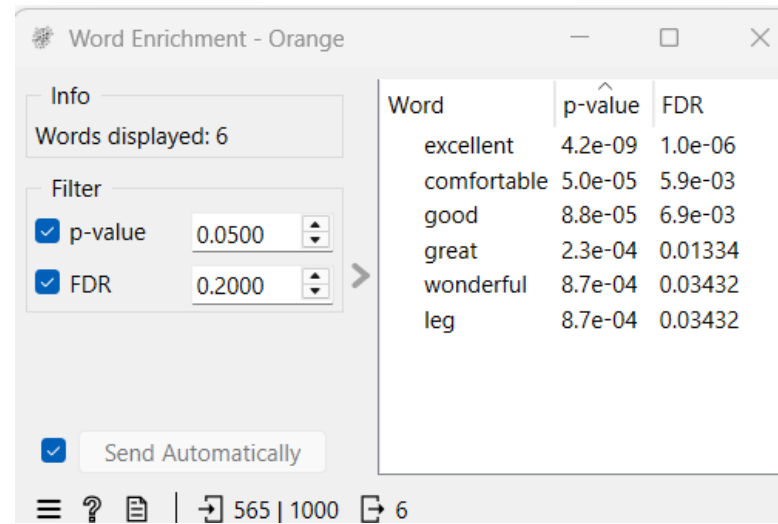
Filter
☒ p-value 0.0500
☒ FDR 0.2000

☒ Send Automatically

Word	p-value	FDR
even	2.9e-05	2.8e-04
still	3.8e-05	3.6e-04
got	4.2e-05	3.8e-04
two	5.1e-05	4.5e-04
think	8.2e-05	6.9e-04
premium	1.1e-04	8.9e-04
one	1.3e-04	1.0e-03
know	1.6e-04	1.2e-03
booked	1.9e-04	1.4e-03
help	2.4e-04	1.7e-03

195 | 1000 92

C2



Word Enrichment - Orange

Info
Words displayed: 6

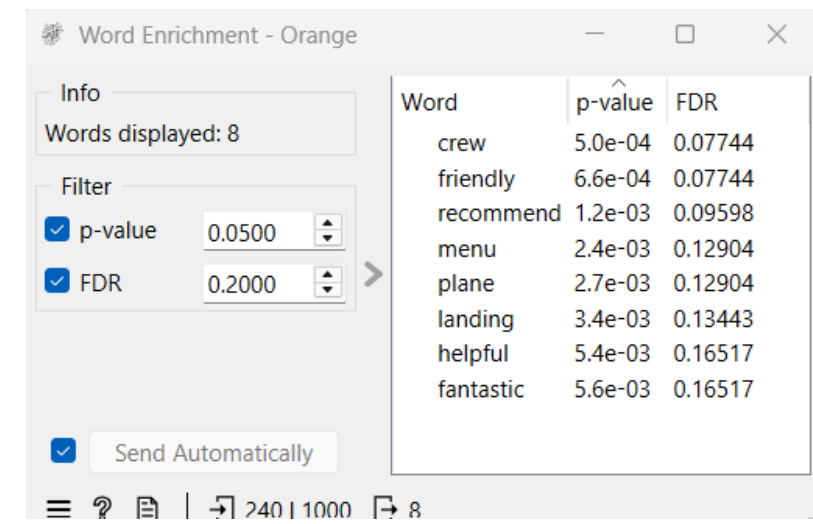
Filter
☒ p-value 0.0500
☒ FDR 0.2000

☒ Send Automatically

Word	p-value	FDR
excellent	4.2e-09	1.0e-06
comfortable	5.0e-05	5.9e-03
good	8.8e-05	6.9e-03
great	2.3e-04	0.01334
wonderful	8.7e-04	0.03432
leg	8.7e-04	0.03432

565 | 1000 6

C3



Word Enrichment - Orange

Info
Words displayed: 8

Filter
☒ p-value 0.0500
☒ FDR 0.2000

☒ Send Automatically

Word	p-value	FDR
crew	5.0e-04	0.07744
friendly	6.6e-04	0.07744
recommend	1.2e-03	0.09598
menu	2.4e-03	0.12904
plane	2.7e-03	0.12904
landing	3.4e-03	0.13443
helpful	5.4e-03	0.16517
fantastic	5.6e-03	0.16517

240 | 1000 8

Insights from Clustering Analysis

C1: Reviews focus broadly on economy class, seats, service, delays, and overall travel experience. The enriched words suggest varied passenger experiences, including booking and premium/ economy services.

C2: Reviews are strongly associated with positive experiences, with enriched words such as excellent, comfortable, good, great, and wonderful, particularly relating to comfort and service.

C3: Reviews mainly highlight crew, food/menu, flights, and landing, with crew and friendly being significantly enriched.

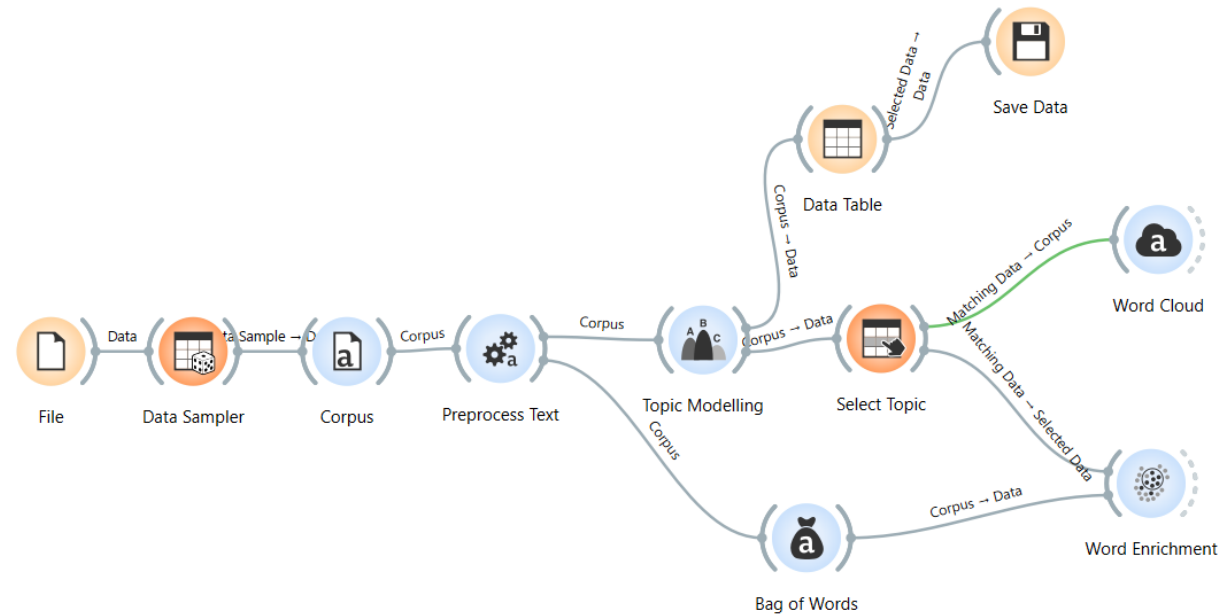
Overall, the clusters reveal three distinct customer experience patterns: general flight and seating experiences (C1), positive comfort and service experiences (C2), and crew and food-related experiences (C3). This shows that customer reviews can be grouped according to different aspects of the Singapore Airlines experience.

What is LDA?

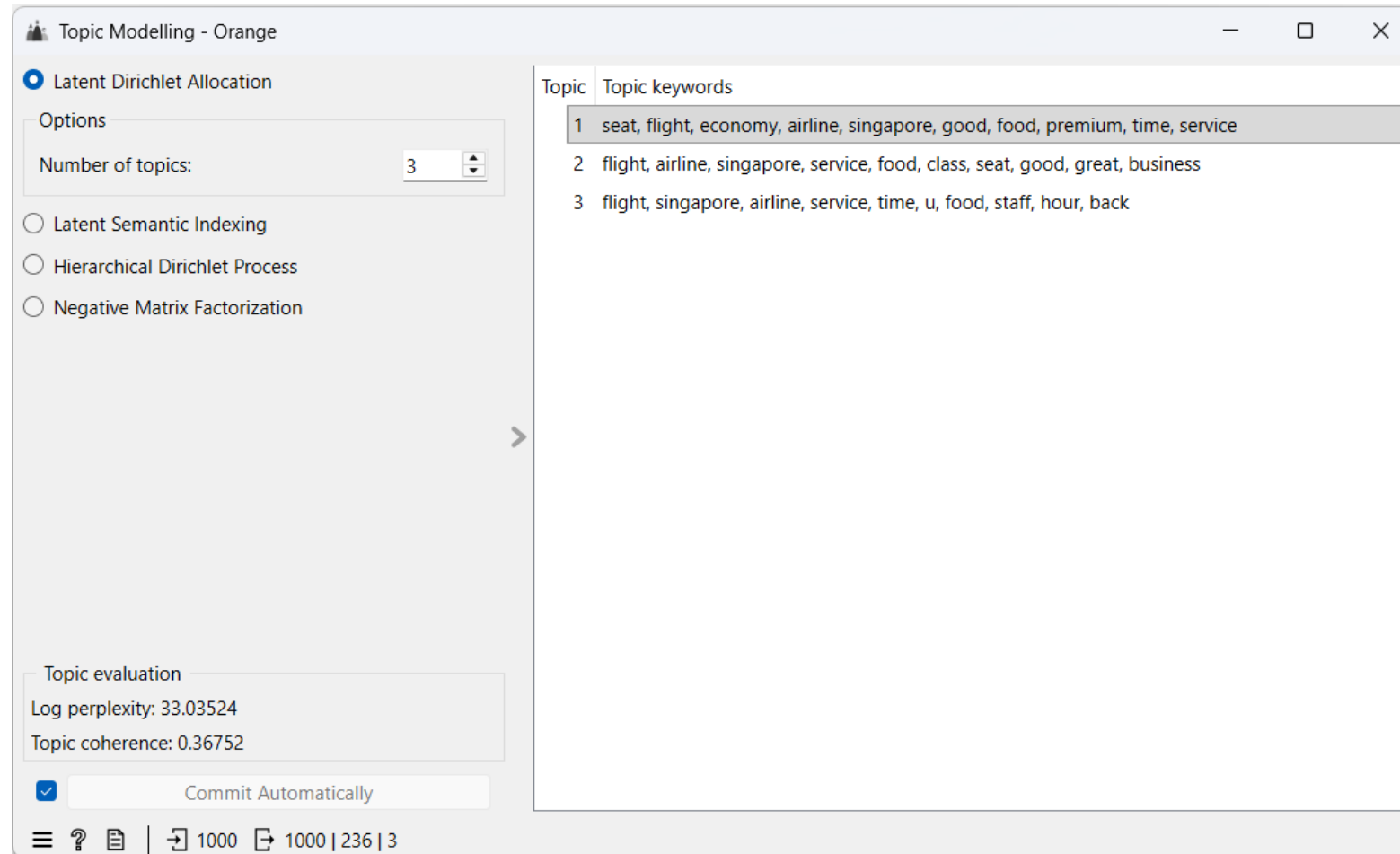
- **LDA** is an **unsupervised topic modelling** technique used to discover hidden topics in a collection of documents.
 - Assumes each document contains a mixture of topics.
 - Each topic is represented by a group of frequently associated words.
 - Assigns probabilities to topics within documents and words within topics.
 - Requires specifying the number of topics.

Topic-Modelling Analysis Pipelines

- The pipeline preprocesses the review text, performs text modelling to identify key topics, and analyses the selected topic using a word cloud and word enrichment.



Topic Modelling (LDA)



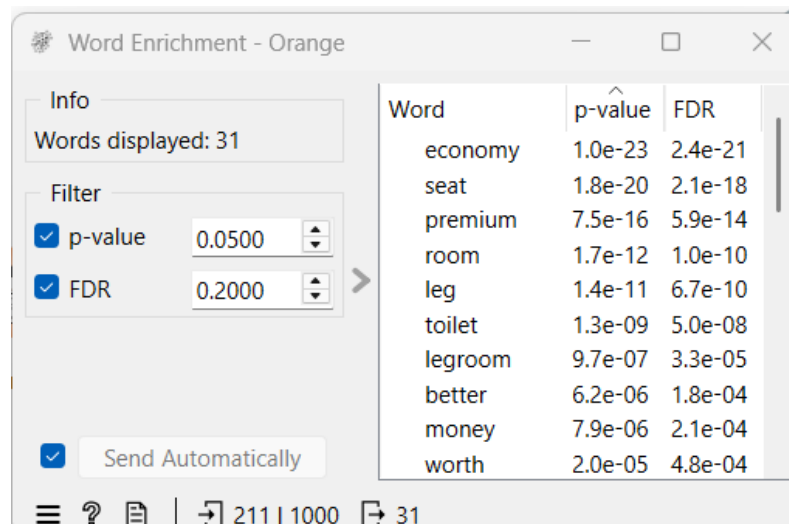
The screenshot displays the 'Topic Modelling - Orange' window. On the left, the 'Latent Dirichlet Allocation' method is selected. The 'Number of topics' is set to 3. Below this, three other methods are listed but not selected: 'Latent Semantic Indexing', 'Hierarchical Dirichlet Process', and 'Negative Matrix Factorization'. At the bottom left, the 'Topic evaluation' section shows 'Log perplexity: 33.03524' and 'Topic coherence: 0.36752'. A 'Commit Automatically' checkbox is checked. On the right, a table lists the extracted topics and their keywords.

Topic	Topic keywords
1	seat, flight, economy, airline, singapore, good, food, premium, time, service
2	flight, airline, singapore, service, food, class, seat, good, great, business
3	flight, singapore, airline, service, time, u, food, staff, hour, back

At the bottom of the window, there is a status bar with icons for help, file, and a progress indicator showing '1000 | 1000 | 236 | 3'.

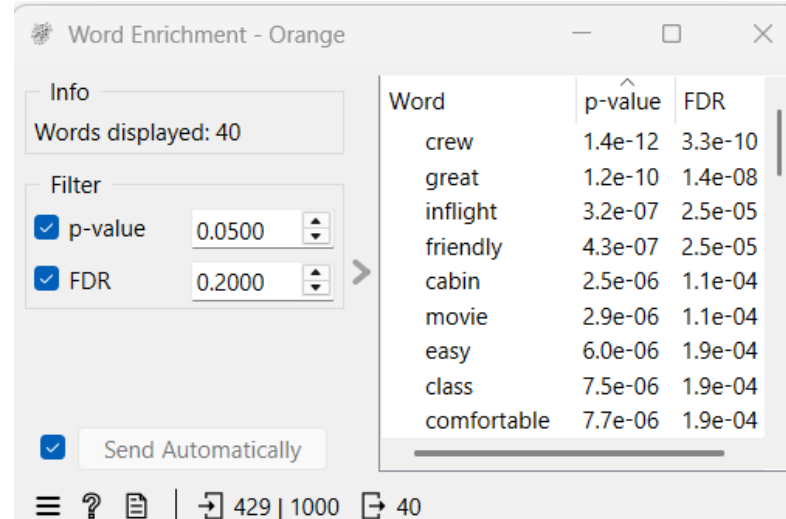
Topic Modelling Word Enrichment

Topic 1



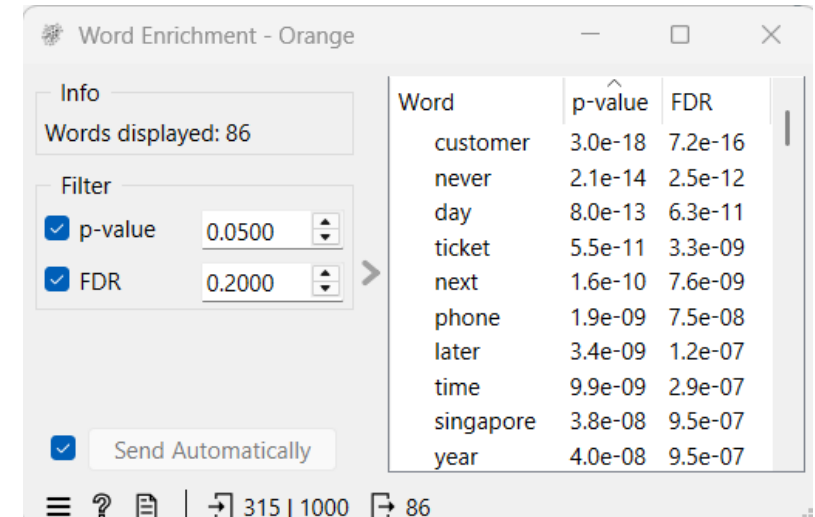
Word	p-value	FDR
economy	1.0e-23	2.4e-21
seat	1.8e-20	2.1e-18
premium	7.5e-16	5.9e-14
room	1.7e-12	1.0e-10
leg	1.4e-11	6.7e-10
toilet	1.3e-09	5.0e-08
legroom	9.7e-07	3.3e-05
better	6.2e-06	1.8e-04
money	7.9e-06	2.1e-04
worth	2.0e-05	4.8e-04

Topic 2



Word	p-value	FDR
crew	1.4e-12	3.3e-10
great	1.2e-10	1.4e-08
inflight	3.2e-07	2.5e-05
friendly	4.3e-07	2.5e-05
cabin	2.5e-06	1.1e-04
movie	2.9e-06	1.1e-04
easy	6.0e-06	1.9e-04
class	7.5e-06	1.9e-04
comfortable	7.7e-06	1.9e-04

Topic 3



Word	p-value	FDR
customer	3.0e-18	7.2e-16
never	2.1e-14	2.5e-12
day	8.0e-13	6.3e-11
ticket	5.5e-11	3.3e-09
next	1.6e-10	7.6e-09
phone	1.9e-09	7.5e-08
later	3.4e-09	1.2e-07
time	9.9e-09	2.9e-07
singapore	3.8e-08	9.5e-07
year	4.0e-08	9.5e-07

Insights from Topic Modelling

Topic 1: Focuses on economy and premium travel experiences, particularly seats, flights, food , legroom, and inflight comfort.

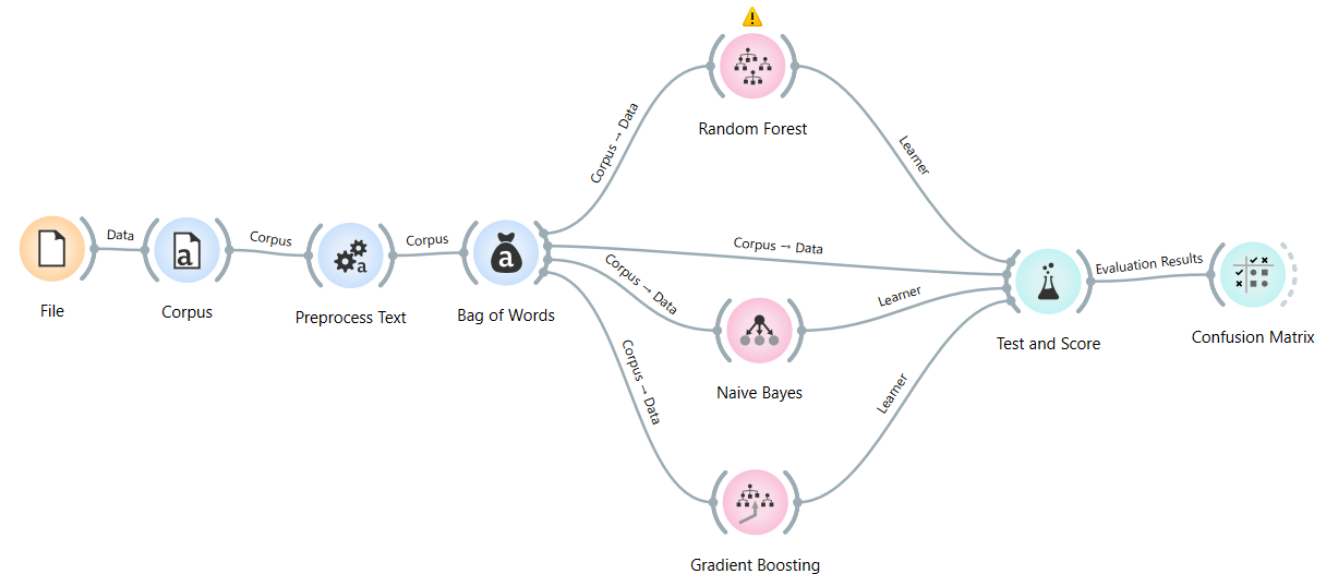
Topic 2: Highlights service quality and cabin experience, with enriched words such as crew, great, friendly, and comfortable, indicating generally positive feedback.

Topic 3: Focuses on customer service and flight operations, including customer interactions, delays, boarding, and overall airline service.

Overall, the topic modelling identifies three main areas of customer feedback: cabin and comfort, service quality, and flight/ customer service experiences. This suggests that these factors are key themes influencing passengers' overall experience with Singapore Airlines.

Text Classification Pipelines

- The pipeline preprocesses the review text, converts it into bag of words features, trains Random Forest, Naïve Bayes, and Gradient Boosting classifiers, and evaluates their performance using cross-validation and a confusion matrix.



Text Classification (Cross-Validation)

Test and Score - Orange

☒ Cross validation

Number of folds: 5

☒ Stratified

☐ Cross validation by feature

☐ Random sampling

Repeat train/test: 10

Training set size: 80 %

☒ Stratified

☐ Leave one out

☐ Test on train data

☐ Test on test data

Evaluation results for target (None, show average over classes)

Model	AUC	CA	F1	Prec	Recall	MCC
Random Forest	0.847	0.792	0.769	0.756	0.792	0.396
Naive Bayes	0.815	0.698	0.728	0.775	0.698	0.359
Gradient Boosting	0.846	0.801	0.785	0.774	0.801	0.439

Compare models by: Area under ROC curve

☐ Negligible diff.: 0.1

	Random Forest	Naive Bayes	Gradient Boosting
Random Forest		0.844	0.565
Naive Bayes	0.156		0.119
Gradient Boosting	0.435	0.881	

Table shows probabilities that the score for the model in the row is higher than that of the model in the column. Small numbers show the probability that the difference is negligible.

Text Classification (Confusion Matrix)

Random Forest

		Predicted			Σ
		negative	neutral	positive	
Actual	negative	11	23	13	47
	neutral	23	25	51	99
	positive	6	13	454	473
	Σ	40	61	518	619

Naïve Bayes

		Predicted			Σ
		negative	neutral	positive	
Actual	negative	25	18	4	47
	neutral	34	37	28	99
	positive	49	54	370	473
Σ		108	109	402	619

Gradient Boosting

		Predicted			Σ
		negative	neutral	positive	
Actual	negative	10	28	9	47
	neutral	20	34	45	99
	positive	4	17	452	473
Σ		34	79	506	619

Insights from Text Classification

- Gradient Boosting performed best overall, achieving the highest classification accuracy (0.801), F1-score (0.785), and MCC (0.439), while Random Forest had the highest AUC (0.847).
- All three models classified positive reviews most accurately. However, negative and neutral reviews were more frequently misclassified, particularly as neutral or positive.
- Naïve Bayes showed the weakest overall performance, with the lowest accuracy (0.698) and AUC (0.815).
- Overall, the results indicate that Gradient Boosting is the most suitable model for classifying sentiment, but the models perform better for positive sentiment than negative or neutral sentiment, suggesting difficulty in distinguishing between these less frequent sentiment categories.

Conclusion



VADER sentiment analysis identified whether reviews expressed positive, neutral or negative opinions, showing that positive experiences were mainly associated with service, comfort, and food, while negative feedback focused more on customer service, tickets, and flight-related issues.



Euclidean hierarchical clustering further grouped similar reviews into distinct customer experience patterns, including general flight and seating experiences, positive comfort and service experiences, and crew and food-related experiences.



LDA topic modelling identified the main discussion themes, such as cabin comfort, service quality, and flight or customer service experiences.



Text classification used these review patterns to predict sentiment categories, with Gradient Boosting achieving the best overall performance, although positive reviews were classified more accurately than neutral and negative reviews due to class imbalance.